Mini Project: 555 Timer

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Abstract

The 555 timer, found under many chip names such as the LM555CN, NE555, TLC555CP, LMC555, and so on, is a timer chip configurable in both monostable and astable formats, both of which are general applications. The chip is an integrated circuit, which allows an incredibly wide variety of applications such as doorbells, sirens, and lights, to be toggled on and off repeatedly over a pre-calculated time period. The time period of the circuit can be calculated via chosen capacitor and resistor values, while the timer actually functions by charging and discharging the capacitors in order to change potential voltage differences and ultimately turn its outputs on and off "periodically". Both monostable, astable, and a specific application (in this case, police lights) will be analyzed and tested.

1. Objectives

To understand the fundamental operating procedure of the 555 IC and analyze the precise nature of its behavior when implemented. Secondarily, to analyze the practical applications of the chip, across multiple configurations.

1.1. Monostable LMC555 IC With 3 Second High Output Pulse

The objective of the first of four tasks within this report is to create a 555 timer circuit in a monostable configuration specifically using a pull-up trigger pin. The circuit, when triggered, will create a 3 second pulse of high output. It is worth noting that 3 seconds is the objective for this task in particular, but is only the value of the circuits time constant in general. The calculations for capacitance and resistance values necessary to achieve 3 seconds will be discussed in the next section. Being that this is a monostable configuration, there is no "automation", and the timer will only be triggered when done so by the user. In LTSpice, this is achieved used a specifically configured voltage to the trigger, pin, and in reality, this will be achieved using a "momentary" button. Once the circuit is triggered, a high output will be observed, for only as long as the capacitor (which for simplicity, will be arbitrarily chosen within reason) is charged to ⅔ or more of the input voltage. As aforementioned, the calculations to find the necessary resistor value, using a chosen capacitor, 3 second time period, and this (⅔) coefficient, will be found in section 2.

1.2. Astable LMC555 IC | 60% Duty Cycle | 2 Second Time Period

The second of four tasks within this report is to configure the 555 timer in its atable mode, with a 2 second time period and a 60% duty cycle. Being that it is now an astable arrangement, there is a level of “autonomy” introduced, as the circuit will more or less trigger itself. The duty cycle refers to the ratio of high level and low level output from the circuit, which for a 2 second time period means that 1.2 seconds of high output and 0.8 seconds of low output can be expected. When executed correctly, this will create a square wave, with a negligible “rise time” (it can be considered very close to 0). This task required the construction of a 555 timer circuit with the trigger pin set to low, which means that the capacitor will similarly charge to ⅔ of the input voltage at a time rate determined by the resistor values, and discharge to ⅓ of the input voltage, at which point another trigger will occur, and the cycle will repeat. In this fashion, “autonomy” is introduced. The calculations for resistor values can also be found in section 2.

1.3. Astable LMC555 IC | 75% Duty Cycle | 1 Second Time Period

The objective of this task, the third of 4, is identical to the previous one with the alteration of the duty cycle and time period. Using now a 1 second time period, and a 75% duty cycle, a high output can be expected for 75% of the time, or, for 0.75 seconds. This means that a low output can be expected for 0.25 seconds. To accomplish this, the same capacitor, but a different set of resistors is used, which will be calculated in section 2. With these changes aside, the same expectations from the 60% duty cycle apply, being a square wave output, autonomous operation, and capacitor charging values between ⅓ and ⅔ for triggering purposes.

1.4. LMC555 IC | Practical Application | Police Lights

The final task in this report is to practically apply the 555 timer IC to create a simulated police or emergency vehicle light circuit. The use of two 555 timers, and 6 total LED diodes (2 row of 3 in parallel), will allow for 3 lights to be on at all times, alternating very rapidly. By charging and discharging the capacitor between ⅓ and ⅔ of the 5V input voltage very quickly, a realistic representation of real world police lights can be achieved. The use of two LC555 IC chips is necessary, to ensure that when configured properly, one is at its high output while the other is at its low output, which creates the oscillatory flashing effect of the LED’s. To further the accuracy of the police light simulation, one of the 3 LED’s in parallel (in each row), can be reversed (swap the anode and cathode), which will effectively put it in sync with the other color of LED’s. This addition was tested, but will not be discussed further in this report.

2. Theory

The 555 timer chip, specifically the LMC555 is essentially a complex voltage divider between the chips Vcc pin and its GND pin, using three different 5KΩ resistors, hence the 555 name. Within its circuitry, there are also reference points set to ⅓ and ⅔ of the input voltage of 5V in this case, which act as thresholds for the 3 internal comparators to monitor voltage across the divider network. It is these comparators that allow adjustable time periods, and open up such a wide array of applications. In order to discuss the remainder of the LMC555's circuitry a diagram is necessary.

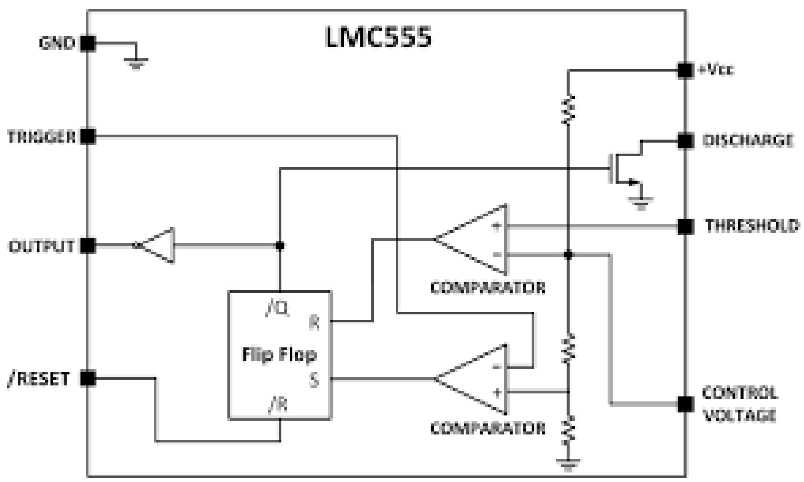

Figure 1: LMC555 IC Internal Circuit Diagram.

The first pin, or pin 1, is the ground pin, which is wired to the negative terminal of the power supply in this case. Pin 2 pictured above, is the "trigger" pin, which is wired to whatever the trigger source may be. In this monostable application, it is wired to a button interrupting a power/ground path, and an in astable configuration it is wired to (in this case), an external divider network, going to the threshold pin. Internally though, it controls the flip flop function through one of the comparators. The output pin is relatively self explanatory, but in practice is connected to load or oscilloscope for measurement/analysis. Pin 5, the control voltage, controls the duration of the pulse at the output and pin 6, the threshold pin, is one of the comparators mentioned earlier. This comparator sets the flip flop to ⅔ of the input voltage (5V). The discharge pin (pin 7), is actually responsible for toggling the high and low output (see the ground node) based on the ⅔ of input voltage threshold, as well as for discharging the capacitor. Finally, Vcc, pin 8, is the voltage input to the circuit.

The observed behavior at the output of the pin (whether it is high or low output), is determined at the flip flop, but more specifically by the comparators connected to the flip flop. These comparators are the ⅔ comparators, which as detailed above, determine the output level based on circuit resistor values and input voltage.

2.1. Monostable LMC555 IC Theory

To create the mono stable 555 timer circuit, the crucial values are the resistor and capacitor values, as well as the desired time constant. The time constant must first be decided, and in this case, the capacitor value as well. For this circuit, a 100μF capacitor will be used, and as defined in section 1, a 3 second time period is desired. To calculate this, equation 1 below is used.

$$\tau = 1.1 * R * C$$

Equation 1: Time Constant Equation for Monostable 555 Timer

When executing testing of this circuit, for diagnostic and assembly purposes, it is critical to understand what each component does, and when its action occurs. This is outlined in the following flow diagram.

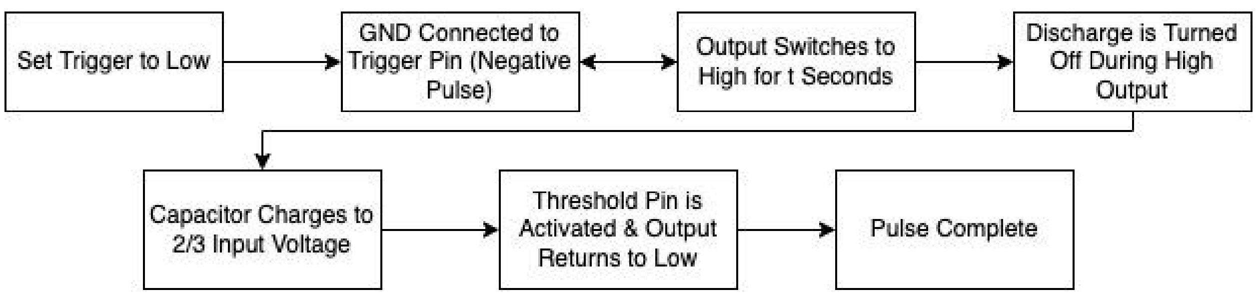

Figure 2: Monostable Configuration General Operation Flow Diagram

2.2. Astable LMC555 IC Theory

As with a monostable application, there are critical formulas to determine resistor and capacitor values for an astable 555 timer circuit. In this case, it is a slightly more involved task, as the circuit will run autonomously until terminated, in a loop. First, the desired time constant must be determined, as well as the value of the timing capacitor to be used and the duty cycle. These values can then be plugged into the following equations to find the required value of R_1 and R_2. It is worth noting, that there are two values for time, one of which is the value of high output based on the duty cycle (τ_1) and the other is the value of low output (τ_2). To solve for each resistor value, first solve τ_2 and then τ_1, based on the duty cycle equation. As was the case for the monostable setup, a 100μF capacitor will be used, and two difference circuits of $\tau = 2s$ with a 60% duty cycle, and $\tau = 1s$ with a 75% duty cycle will be analyzed.

$$DC = \left(\frac{High\ Voltage}{\tau}\right) * 100$$

Equation 2: Duty Cycle calculation

$$\tau_1 = 0.693 * (R_2 + R_1) * C_1$$

Equation 3: Time Constant 1 Equation for Astable 555 Timer

$$\tau_2 = 0.693 * R_2 * C_1$$

Equation 4: Time Constant 2 Equation for Astable 555 Timer

With calculations all sorted, once resistor and capacitor values have been determined and it is time to build the circuit, it is first very important to understand exactly how the astable configuration for a 555 timer functions. Pins 2 and 6, the trigger and threshold pins are connected in this setup, allowing for a loop to occur, with high and low voltage being observed every time the capacitor charges and discharges to ⅓ and ⅔ of the input voltage. The capacitor charges through two resistors this time, and discharges through one, as there must be two paths of travel to allow for continuous repetition of charging and discharging / high and low output. The operational flow diagram below outlines this process. (Intermediate steps, such as discharging during charging have been eliminated for simplicity, see figure 2 for detailed outline).

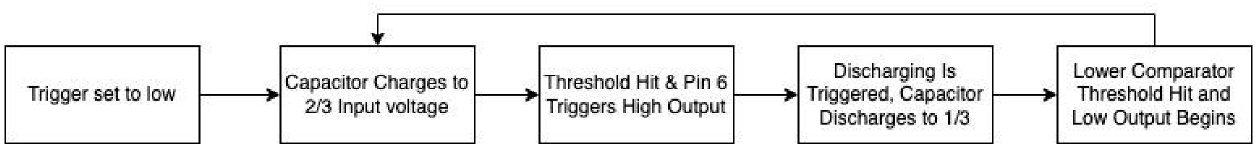

Figure 3: Astable Configuration General Operation Flow Diagram

3. Procedure

The general procedure for each of the four tasks covered within this report, is to first calculate the necessary R and C values for each circuit, and then construct the circuit in LTSpice. Since a real life application cannot be always directly implemented into spice, it is then important to find a modification that will serve its purpose for simulation. For example, although a button does not exist in Spice, using a voltage source with advanced configurations for PULSE, will suffice to trigger the monostable circuit. Once this has been accomplished, simulating the circuit to ensure the output is as expected is the next step. Following this, constructing the same circuit on a breadboard and testing it for a similar readout/function as the spice simulation is the final step, which ultimately increases understanding of the 555 timer.

3.1. Monostable LMC555 IC Procedure

The monostable circuit as discussed previously, is designed to be manually triggered and produce a single pulse, until manually triggered again. The desired time period for this circuit is 3 seconds, and the value of C_2 is predetermined by the circuits data sheet to be 0.1μF. C_1 in this case will be chosen at 100μF.

$$3 = 1.1 * R * 100\mu F$$

Given this value and the 3 second time period, equation 1 can be used to calculate the resistor value at 27.27KΩ. Since this is not a readily available resistor value, and is even higher than using two 10KΩ potentiometers (at which point fine tuning becomes finicky), a 56KΩ and a 47KΩ resistor will be used in parallel, falling just short at 25.55KΩ. Please note that this discrepancy is the source of error in the time period found in section 4 (results)for the monostable configuration, as a 3 second period is expected, but a 3.17 second time period is observed.

Next, a spice circuit is created for simulation purposes, the simulation can be found in section 4.

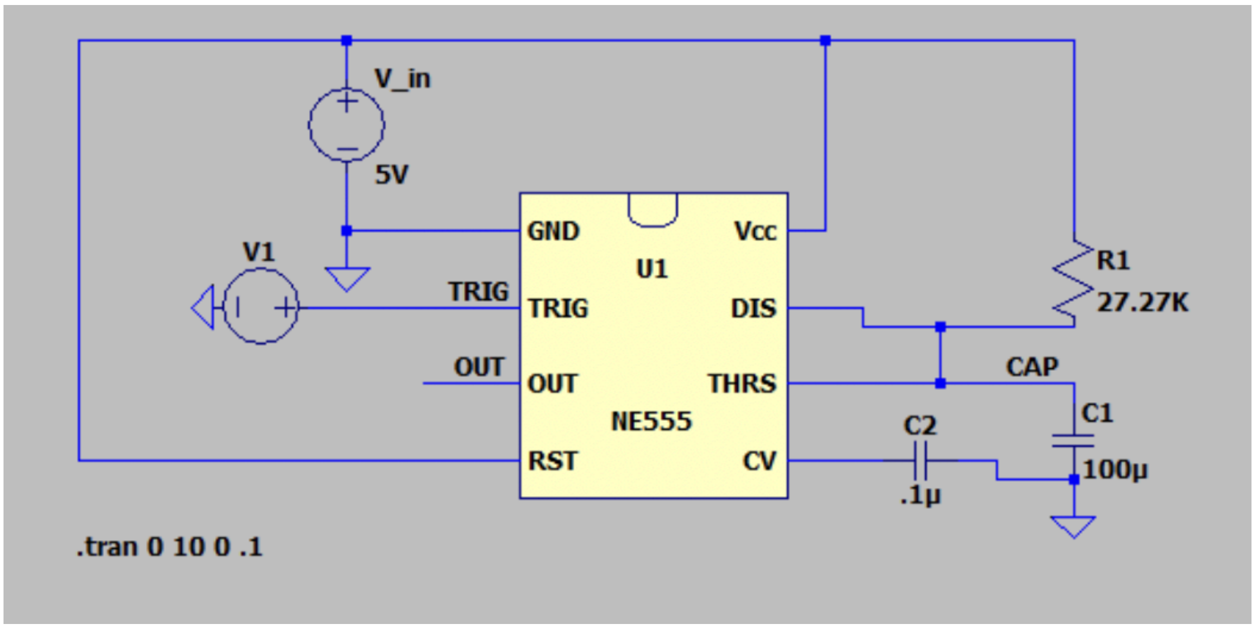

Figure 4: Spice Circuit Schematic for Monostable LMC555 with 3 Second Time Period

Finally, the circuit will be built on a breadboard and analyzed using an oscilloscope measuring output from pin 3, the results of which can also be found in section 4.

3.2. Astable LMC555 IC | 60% Duty Cycle | 2 Second Time Period

As opposed to the single pulse operation of the monostable circuit, the astable circuit is designed to trigger itself using the threshold pin and repeat its operation until input voltage is terminated. Given 60% duty cycle, and a 2 second time period, it is quickly calculated that τ_1 = 1.2s and τ_2= 0.8 seconds. In a similar manner to a monostable circuit, a non-decoupling capacitor value must be chosen, which will also be 100μF for this circuit. Where the application differs though, is that the calculation for resistance values affects both charge *and* discharge times, which matters because their sum must equal the time period itself. While this is not a crucial component of the calculations, it is a good sanity check .

Beginning with R_2, using equation 4, and the values of τ_2 = 0.8 & C_1=100µF, R_2 can be calculated at 11.544KΩ.

$$0.8s = 0.693 * R_2 * 100\mu F$$

Once again, this is not a value that can be directly implemented using a single resistor, so two potentiometers in series will be used. In order to calculate R_1, equation 3 can be used, in combination with the previously stated/calculated values for τ_2, C_1, and R_2, finding that R_1=5.772KΩ.

$$1.2s = 0.693 * (11.544K\Omega + R_1) * 100\mu F$$

A potentiometer is used for this R_2 as well. The use of potentiometers allow for a near-exact real circuit to be built, eliminating as much anticipated deviation from expected values as possible. Any remaining error, is likely the cause of imperfect potentiometer adjustment, internal resistances, and component tolerances.

Next a circuit schematic is created in LTSpice, where it is found that this time around, far less modification to account for the lack of a button or real life components is necessary. This is because this circuit connects its trigger to its threshold, and does not require manual input. The results of the simulation for the circuit in figure 5 can be found in section 4 (results).

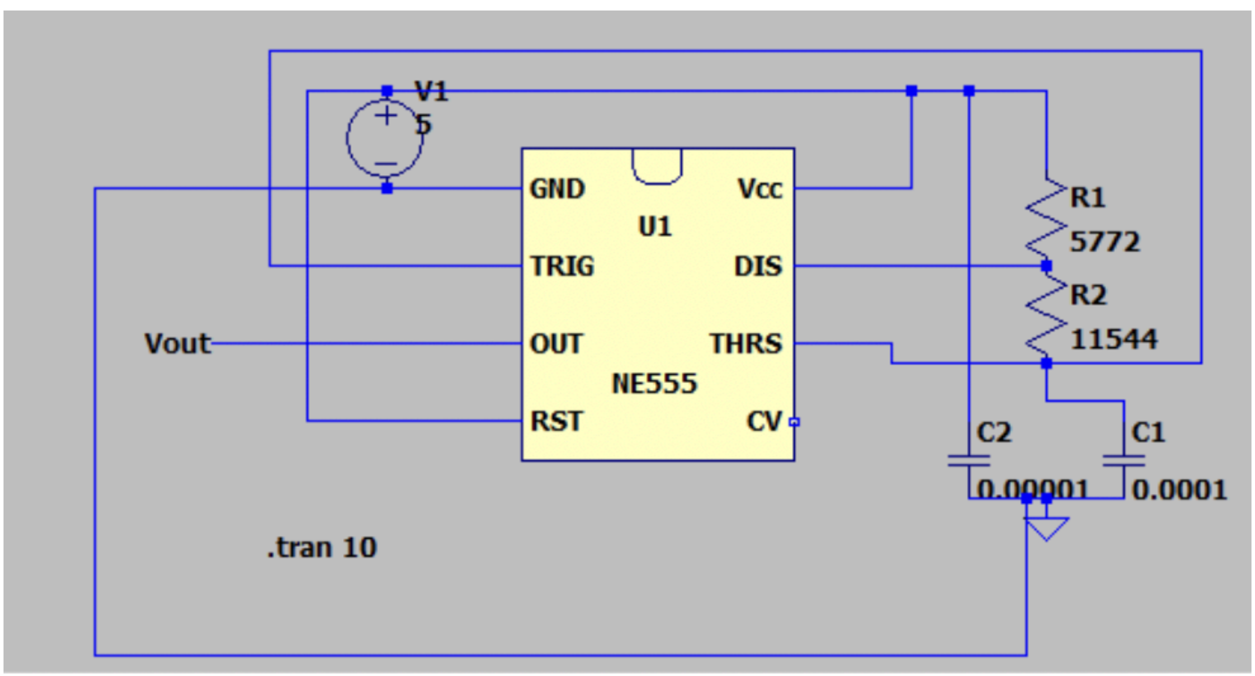

Figure 5: Spice Circuit Schematic for Astable LMC555 with 2 Second Time period and 60% Duty Cycle

3.3. Astable LMC555 IC | 75% Duty Cycle | 1 Second Time Period

The procedure for an astable configuration with a 75% duty cycle and 1 second time period is nearly identical, with a difference observed in the calculated resistance values. With that said, the non-decoupling capacitor value is held the same, and the alterations arrise entirely from the altered values of τ_1 & τ_2. With a 1 second time period and 75% duty cycle, the calculation becomes even clearer, as τ_1 (charge time) will be equal to 75% of the entire time period, which in this case is 0.75s. Similarly, τ_2 (discharge time) will be equal to "the rest", or $\frac{(100-75)}{100} * 1$ which is 0.25s. With these values established, the capacitor value held stable at C_1 =100μF calculating R_1 and R_2 follows the exact same procedure.

Beginning with R_2, using equation 4, and the values of τ_2 = 0.25 & C_1=100μF, R_2 can be calculated at 3.607KΩ.

$$0.25s = 0.693 * R_2 * 100\mu F$$

With R_2 and the value of C_1 & τ_1 we can calculate R_1 to be 7.215KΩ.

$$0.75s = 0.693 * (3.607K\Omega + R_1) * 100\mu F$$

Since neither of these values are directly available in the form of a single resistor, potentiometers will be used for this circuit as well. Given that the circuit diagram depicts 2 resistors in series (acting as the divider), for both duty cycles of the astable configuration, for the 60% duty cycle with one resistors value exceeding 10KΩ, 3 total potentiometers were required. For the 75% duty cycle circuit, being that both resistance values are under 10KΩ, only two potentiometers are required and the circuit is slightly simplified.

The next step is to construct a circuit diagram in LTSpice using these newly calculated values for each resistor and C_1. The circuit is then simulated to ensure the behavior of the circuit is as expected, and its construction on a breadboard shouldnt pose any issues or errors. The result of this simulation can be found in section 4 (results).

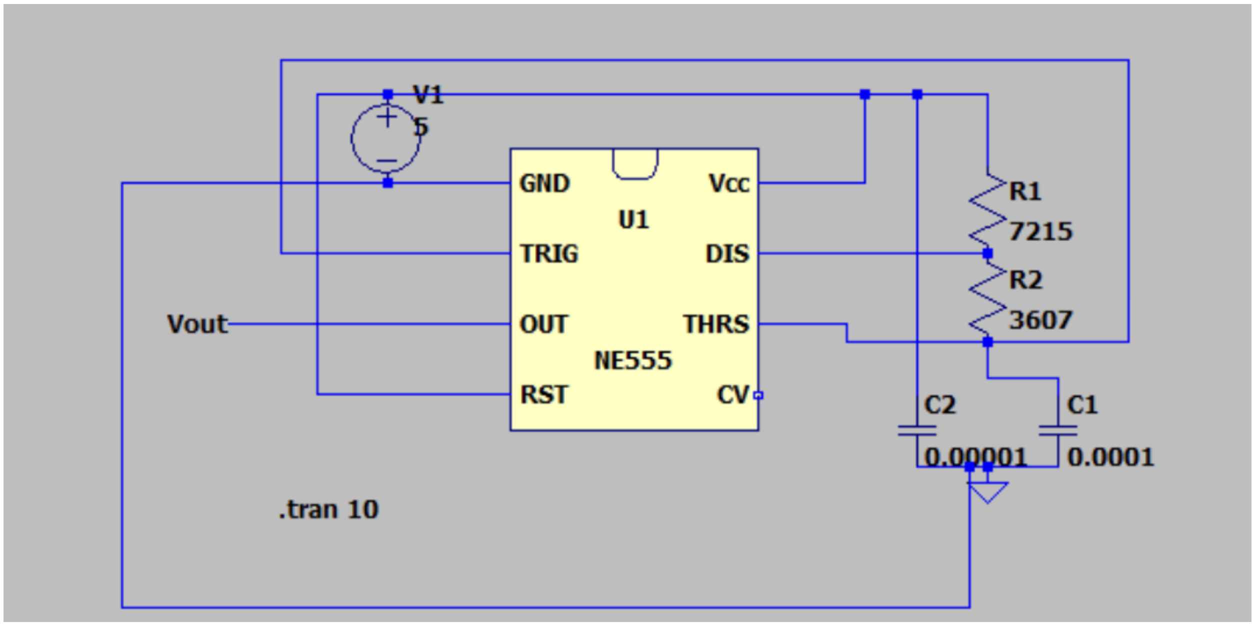

Figure 6: Spice Circuit Schematic for Astable LMC555 with 1 Second Time period and 75% Duty Cycle

Finally, the last procedural step for this particular circuit is to construct it on a breadboard, and analyze its behavior using an oscilloscope at the output pin, ensuring that the output is the same or similar to that of the simulation. This will all be discussed further in section 4.

3.4. Police / Emergency Vehicle Lights | LMC555 IC

The calculations for the police lights circuit are less intensive than those of the previous 3 circuits, as they are provided via the project documentation found on elonics.org. With that said, the complexity of the circuit itself is far higher than that of the monostable or astable configurations. Since the resistor and capacitor values from this reference circuit can be implemented directly, there are no calculations and the first step of the general procedure can by skipped. The inspiration circuit uses two different 555 timers, which control “eachothers” respective group of 3 LEDs. The circuit is configured in such a way that only one output is on at a time, so when the output of the first 555 timer is on, only the first group of 3 LED diodes can turn on, as their anode will receive a negative signal from the *second* 555 timer that is off. It is important to note that both 555 timers are configured in their astable mode. This circuit can be found in figure 7.

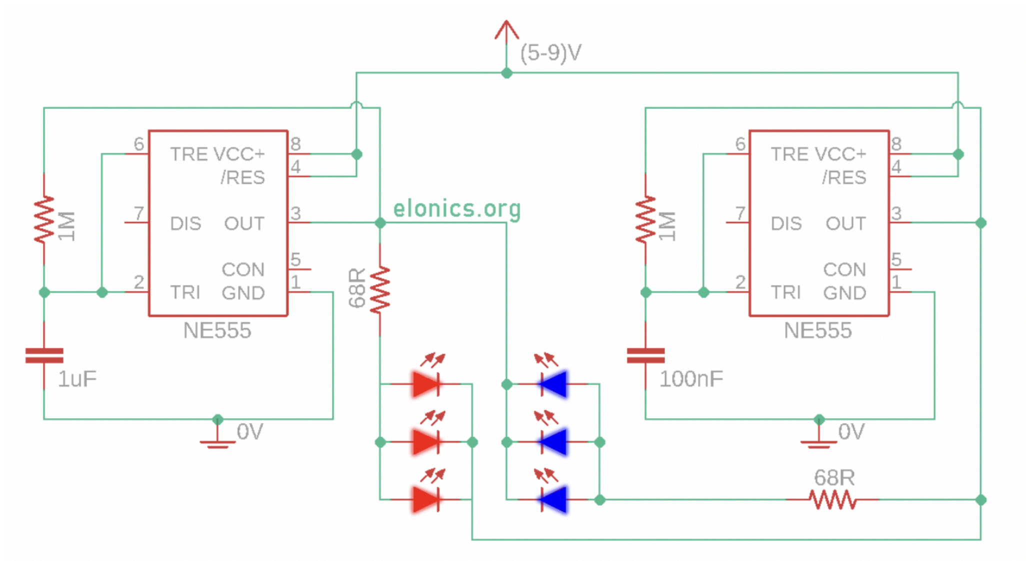

Figure 7: Police Lights LED Flasher Circuit Diagram from Elonics. Org

This concept was expanded upon and tweaked though, in an attempt to increase the realism of the simulation, there needed to be a short (nearly imperceivable) pause in both lights in between each flash cycle, essentially changing the duty cycle of each 555 timer. Additionally, to create the chaotic nature of police lights, as alluded to earlier, reversing the anode and cathode of one led in each row of parallel diodes, essentially syncs it with the other row of LED's, as its cathode now receives voltage from the opposing 555 timer, instead of its anode. This adapted circuit, and the second step of general procedure, can be found in figure 8.

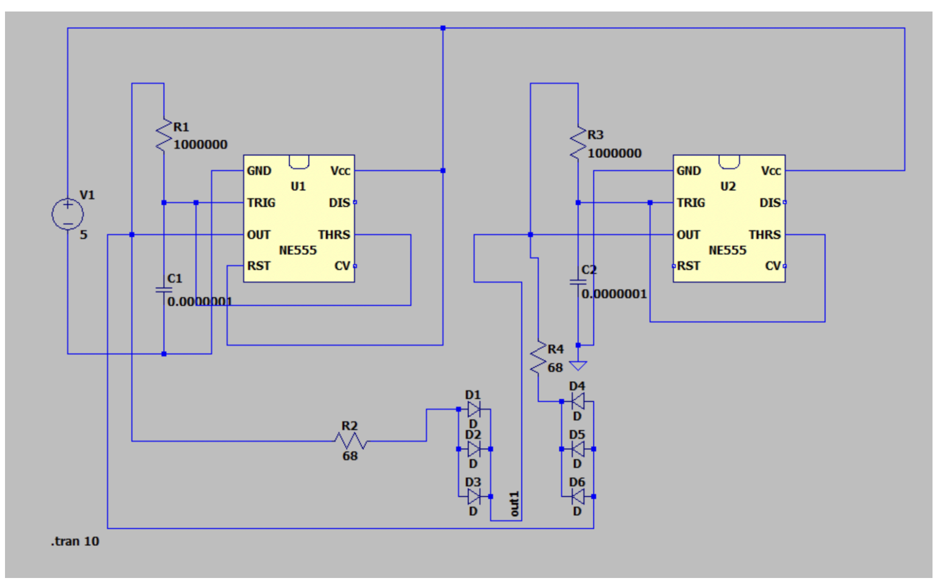

Figure 8: LTSpice Circuit Schematic for Police & Emergency Light Flasher's

With this circuit, a simulation was run to verify functionality, and general output, which should look like a square wave for both outputs, with a short dead period as just described, where both LED's are off momentarily. When both rows of LED's are off, it is because there is 0 potential voltage difference, which will be further discussed in section 4.

Finally, the circuit is built on a breadboard and tested for functionality, with output being measured both at one group of LED's and the other. The reversal of one diode is not percievable on the oscilloscope, as only one LED per row was measured, but it is worth noting that even if the 180 degree rotated diode were measured, its output would simply match that of the other rows.

4. Results

The final two steps of the general procedure outlined in section 3, were a spice simulation of each circuit, and a similarly behaving real circuit, confirmed by oscilloscope readings. These can be found for each circuit with brief analysis below.

4.1. Monostable LMC555 IC With 3 Second Pulse

Once the circuit diagram found in figure 4 was constructed, and lack of real world components was accounted for, the circuit was simulated producing the following results.

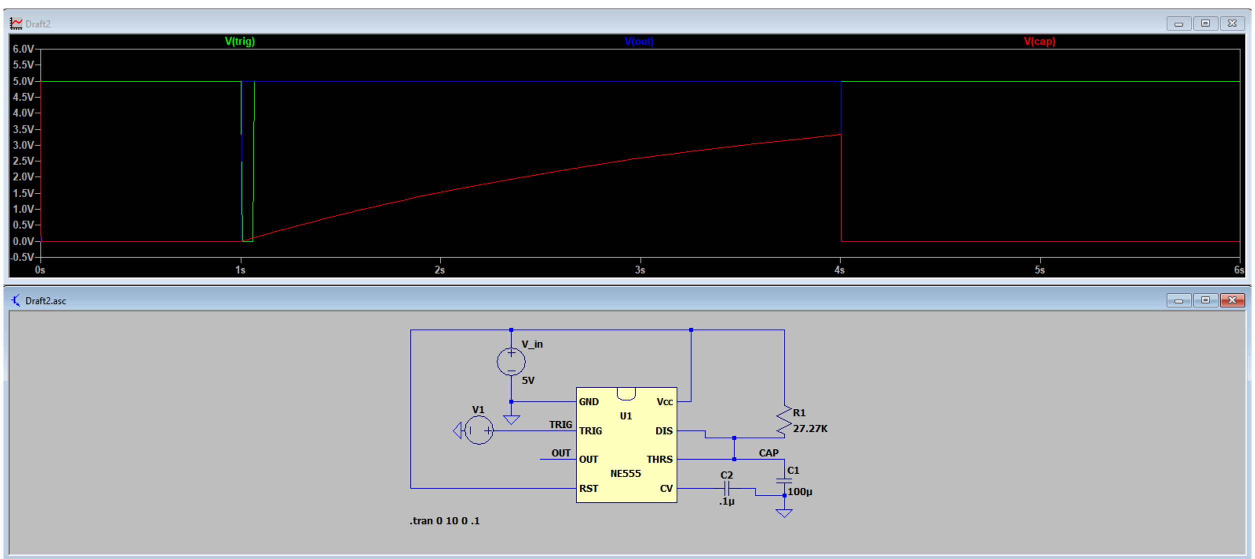

Figure 9: LTSpice Simulation of Monostable LMC555 Circuit With 3 Second Pulse

Although as many points of measurement were not taken on the real circuit, it was pertinent to ensure each component of the circuit was behaving as expected. This circuit was designed with a pull-up trigger, as can be seen from the plot of V_{trig} and V_{out}. The red line indicates the capacitor voltage, the key moment being when the "button" (in this case a PULSE voltage) is pressed. It is ascertainable that V_{trig} momentarily dips down and triggers the circuit “pulling up” the V_{out} value. The duration of this lasts until V_{cap} reaches ⅔ of the applied 5V, at which point the threshold is hit, and the 555 timer resets. The outcome during simulation, represents a nearly perfect 3 second time period. This provided confidence in constructing the circuit using the values for R and C calculated and simulated with .

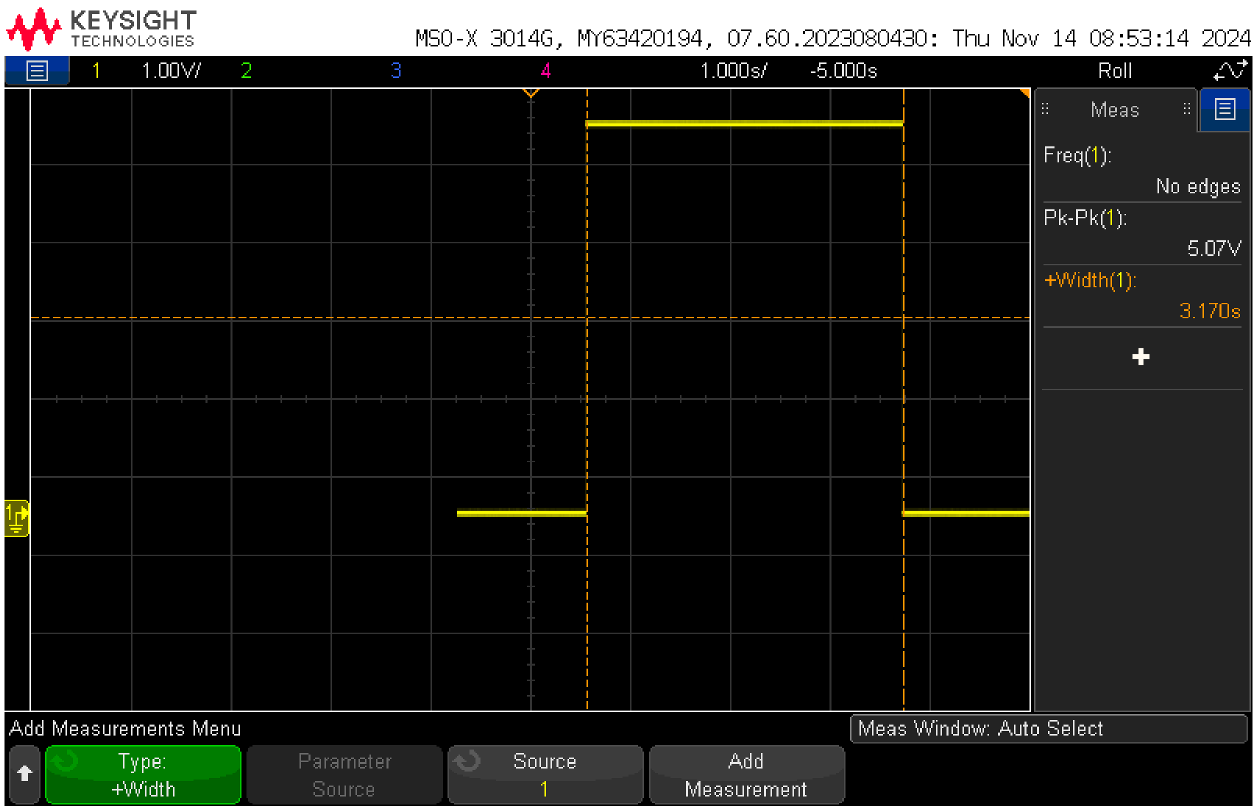

Figure 10: Oscilloscope Measurment of Monostable LMC555 Circuit With 3 Second Pulse

During actual testing a slightly higher measured value was *expected*, as discussed in section 3.1, due to a lower resistance value than 27.27KΩ, of 25.55KΩ. This is exactly what was observed, with a a theoretical time period of 3s and an observed time period of 3.170s. All other things considered though, the real circuit behaved exactly as anticipated and as simulated. With a percent error of:

$$|\frac{3.170-3}{3.170}| * 100 = 5.36\%$$

Combined with the anticipation of a slightly higher result, this circuit performed as expected without any cause fo concern or error.

4.2. Astable LMC555 IC With 2 Second Time period and 60% Duty Cycle

After constructing the circuit found in figure 5, it was simulated in LT spice to verify the square wave behavior anticipated, with a 2 second time period, and high output periods of 1.2 seconds with low output periods of 0.8 seconds. The simulation was run for 10 seconds to verify consistency upon cycle repetitions.

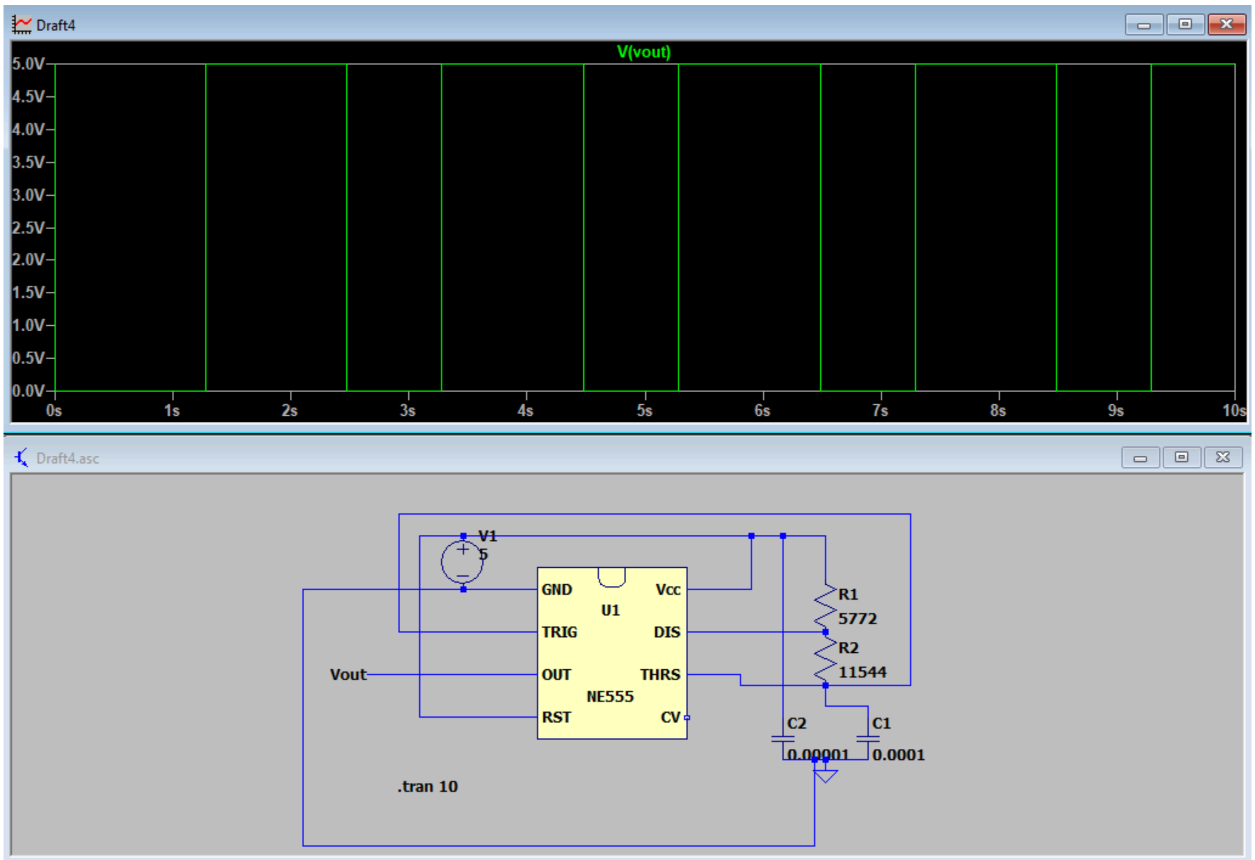

Figure 11: LTSpice Simulation of Astable LMC555 Circuit | 2 Second Time Period | 60% Duty Cycle

The result of this simulation verified the calculated resistor values of 5.772KΩ and 11.544KΩ to create a full time period of 2 seconds with a τ_1 of 1.2s and τ_2 of 0.8s. This installed confidence to create the real circuit and test it using an oscilloscope.

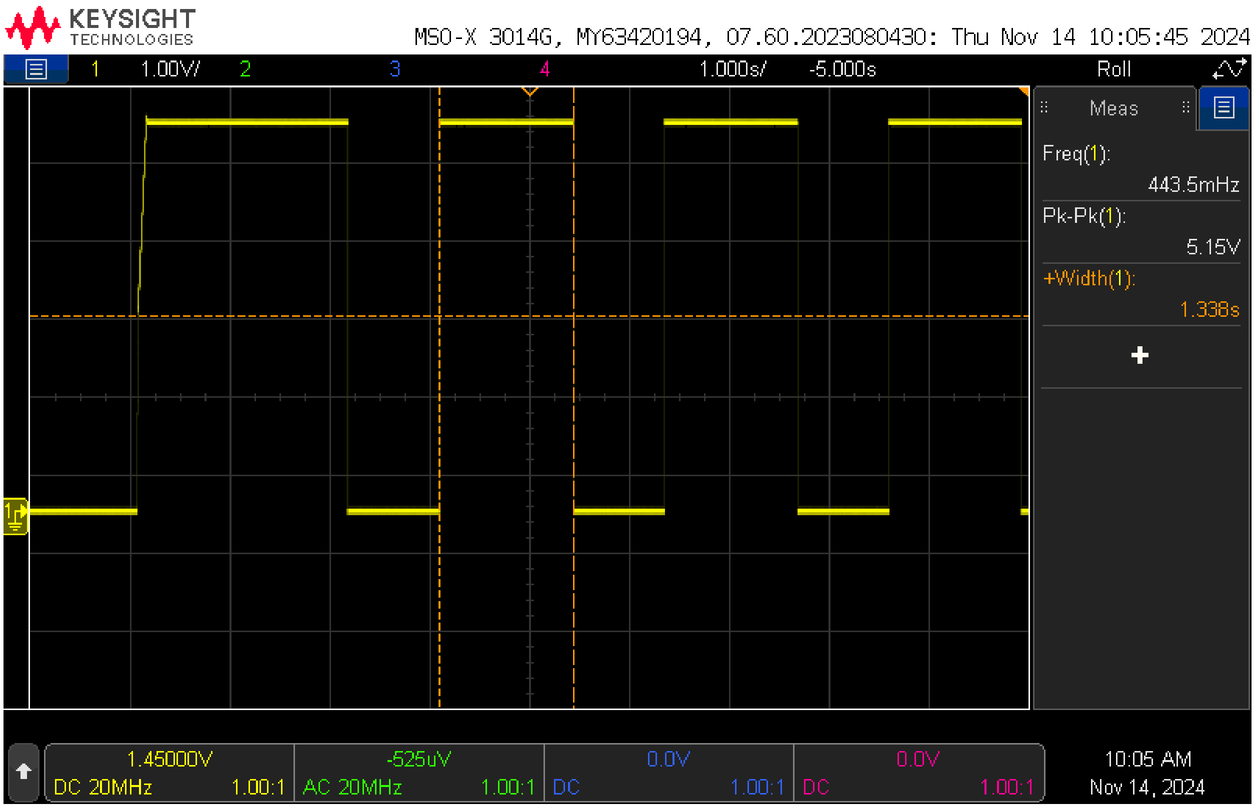

Figure 12: Oscilloscope Measurment of Astable LMC555 Circuit | 2 Second Time Period | 60% Duty Cycle

As was the case with the spice simulation the real circuit demonstrated values extremely close to thos expected and calculated. With an anticipated time period of 2 seconds, and a high output period of 1.2 seconds, a 1.338 second high output period indicates the circuit was built correctly and is functioning as intended. This produces an error of:

$$\left|\frac{1.338-1.2}{1.338}\right| * 100 = 10.3\%$$

While this value can be considered “alarmingly” high, it is once again due to slightly imperfect resistor values, as multiple potentiometers were used during this circuit testing. These particular potentiometers can vary by hundreds of ohms with a bump in the table, so this percent error can be attributed to user error.

Under the assumption that time period itself is relatively exact to 2 seconds, as it demonstrated itself to be in further testing, and the capacitor value has not changed, a duty cycle of 72.5% can be calculated using equation 2:

$$72.5 = \left(\frac{1.45V}{2s}\right) * 100$$

The discrepancy in both duty cycle and high output time period corroborate eachother, indicating that the source of error was in fact the resistor values. All in all, this configuration and testing task do not pose any glaring concerns, as the square wave is output as expected, and the math to inconsistencies checks out.

4.3. Astable LMC555 IC With 1 Second Time period and 75% Duty Cycle

Using the circuit schematic found in figure 6, an LTSpice Simulation was run to verify the calculated resistor and capacitor values, before building the real circuit and implementing these values. It was also run to test functionality and output behavior, but required very little modification to account for lack of components, allowing it to be nearly directly transferable to a breadboard.

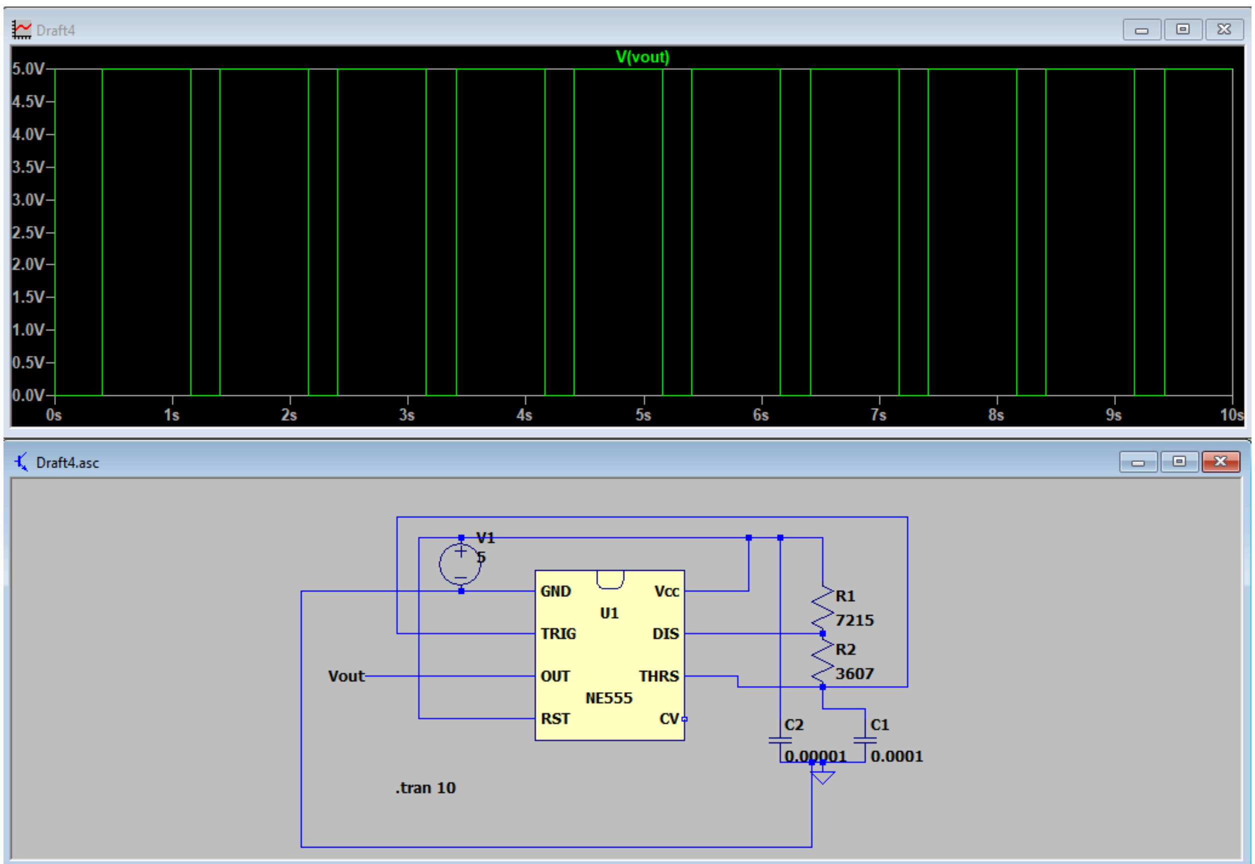

Figure 13: LTSpice Simulation of Astable LMC555 Circuit | 1 Second Time Period | 75% Duty Cycle

This simulation was also run for 10 seconds in an attempt to make sure there would be no inconsistencies over greater time periods or upon repetition. The circuit behaved exactly as expected, producing a square wave with a 1 second time period, and 0.75s high output (with 0.25 second low output). This allowed the real circuit to be built and tested on a breadboard, using an oscilloscop at the output (pin 3).

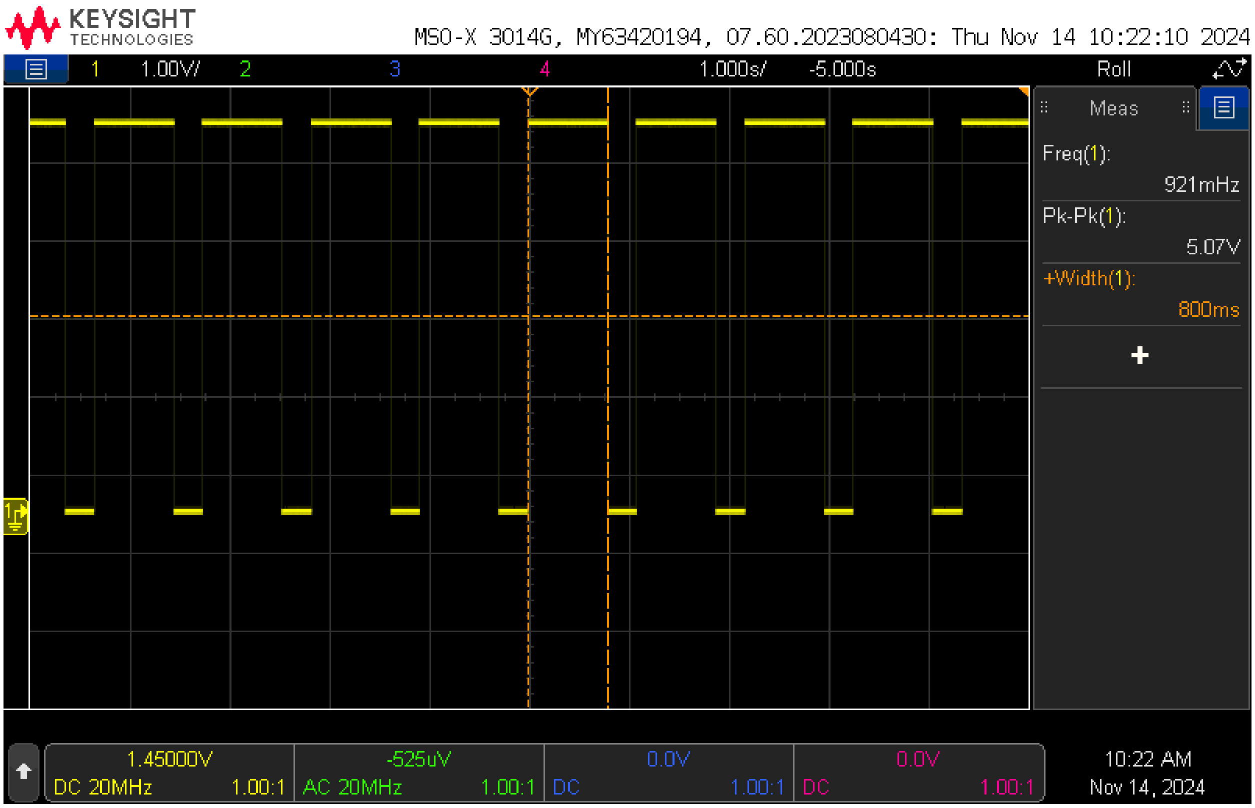

Figure 14: Oscilloscope Measurment of Astable LMC555 Circuit | 1 Second Time Period | 75% Duty Cycle

Upon testing the real circuit, almost the exact behavior expected was observed. While a 0.75s high output and a 0.25 second low output were the theoretical values for the duty cycle and time period, as can be surmised from figure 14, an 800ms, or 0.8s high output value was obtained. This yields a percent error of:

$$\left|\frac{0.8-0.75}{0.8}\right| * 100 = 6.25\%$$

This value is well within the range of acceptable, adn the circuit performed as expected. Assuming there are no internal inconsistencies with internal resistance or poor capacitor values, any and all discrepancy must come from slightly inaccurate resistance values when using potentiometers instead of resistors. Using the duty cycle equation backwards, it can be calculated that the time period during this task was 1.93 seconds. Since changing the potentiometer value changed the time period of the cycle, this value makes sense, and yields a perfectly acceptable percent error of:

$$\left|\frac{2-1.93}{2}\right| * 100 = 3.5\%$$

With this in mind this task was a success, and all measured values are to be expected given a true 1.93 second time period.

4.4. Police / Emergency Vehicle Lights | LMC555 IC

It was discussed previously that the approach for both procedure and data analysis of this circuit varied slightly from the monostable and astable circuits. With that said, when testing the spice simulation for the circuit, a square wave from each output intersecting eachother in an alternating fashion was the expectation during simulation. It is worth noting that the configuration of the simulation window would have been better suited with higher bounds.

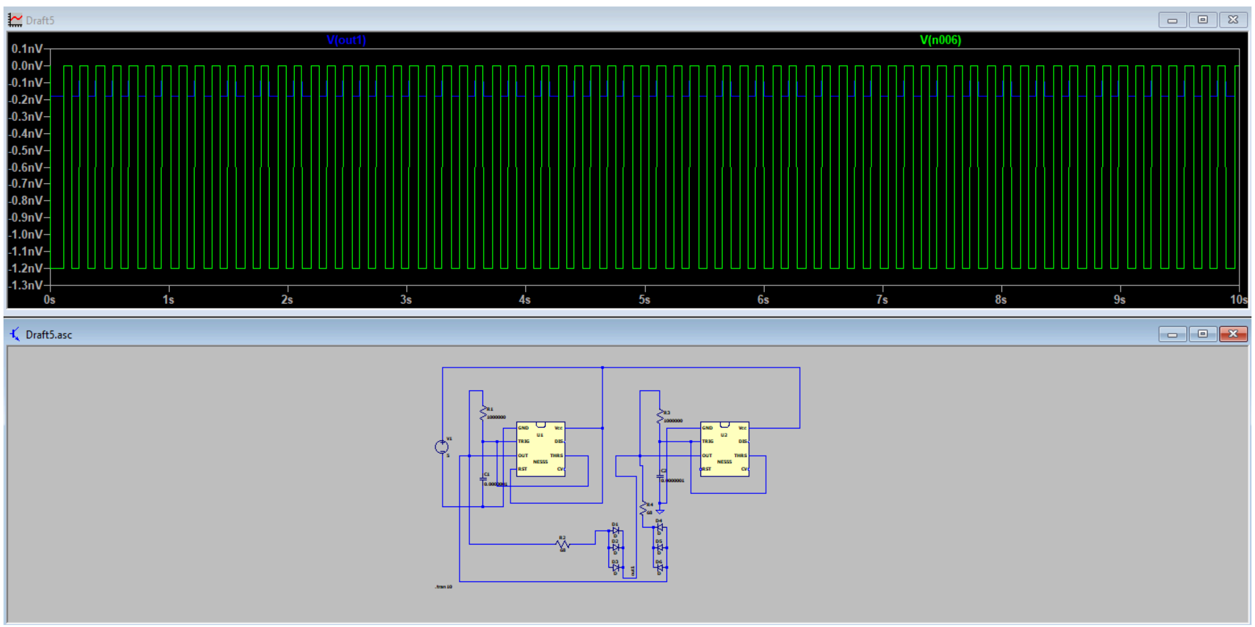

Figure 15: LTSpice Simulation of Police Lights Astable Dual Circuit

This simulation did not provide the exact intended output, but its behavior out of both chips was confirmation enough that the circuit had been created successfully. The lack of accurate representation likely has to do with a missing direct path to ground and poor spice formatting. With this, the actual circuit was built and tested using an oscilloscope on one LED of each row of parallel LED’s.

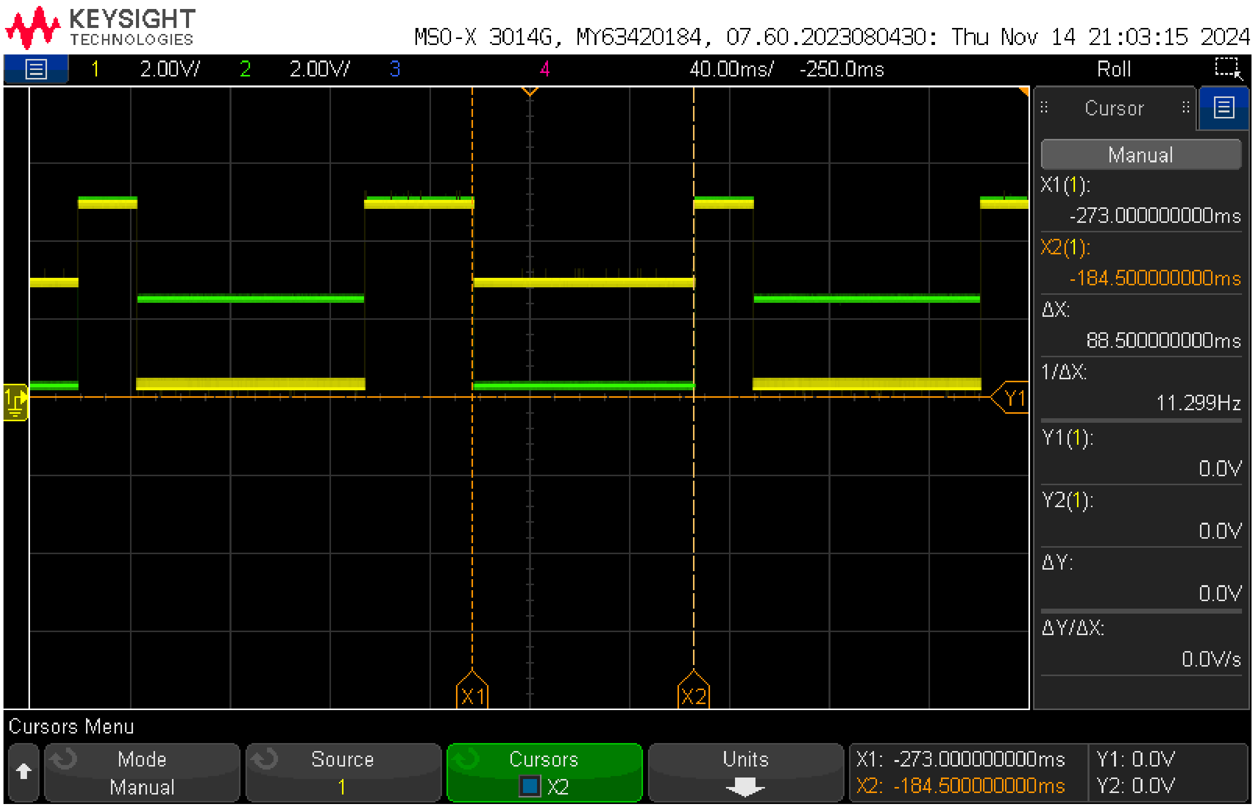

Figure 15: Oscilloscope Measurment of Police Lights Astable Dual Circuit

This oscilloscope snapshot captures 1 of 3 pulses within the circuit that happen every time period. One pulse's duration is 89 ms, with its opposite pulse lasting an equal amount of time with the opposite LED engaged, indicating that the total period is 218 ± 5 ms. The flashing of each LED happens extremely rapidly, with the previously mentioned short off time of both LED's occurring when both voltages are high and match eachother.

Both LED's are off extremely briefly when both timers are high. When this occurs, the potential difference is near to 0, meaning that neither LED can actually turn on, leaving them both off. When timer 1 is high, and time 2 is low, timer 2's lights are on, because timer 2's potential difference is + and timer 1's potential difference is negative. With that, the negative from timer one, is what connects to the annode of timer 2's LED's, allowing them to turn on. The exact reciprocal of this applies when timer 1's LED's are on.

Such rapid toggling between one row of LED's or the other, is only made possible with the use of two 555 timers, because at any given time, both timers are contributing to powering the LED.

This particular design interested me because I've always been fascinated with LED's and unique methods of controlling them. I also have a strong background and interest in automotive, so comprehending how this may be implemented into a vehicle was of interest to me. I have created LED music equalizers, worked with LED'S to use color and relate it to the mood it induces, and this project seemed right up my alley. This application is relatively efficient as it is, and without removing the 555 timer itself, introducing a more advanced single chip or a small pcb computer such as an arduino, I dont see many methods of improving it. With that said, if those *were* options, that is how I would improve the circuit. Its pros are that it can be made extremely small, and is simple to produce in the grand scheme of things. It is efficient, and presumably reliable but tis cons are that for high-intensity applications such as this one, even on a larger scale, more and more 555 timers will be required to expand upon the function of the application.

A video of this particular circuits operation can be found here:
https://drive.google.com/file/d/18e-iWPcoHTpQ9ML2sTLbDc4K5l8ipt2f/view?usp=sharing

5. Conclusions

To first answer the questions posed in the project document, “normally on” and “normally off” refer to the output pin of the LCM555 timer chip, insinuating that the chip will either allow voltage to flow, or it will block it, based on the base state it is in (when it is not in a cycle). As for using a potentiometer instead of a fixed resistance, the effect can be found in the results of this report. Using a potentiometer will allow the user to change the time period of the cycle live, without having to cut the oscilloscope readin by breaking the circuit to swap resistors. The con of doing this, is that potentiometers are often not as precise as a standalone resistor, which will throw off values if a specific output is desired. It would be helpful though, in situations in which you would like a variable time period, in a doorbell with multiple chimes (and durations) for example.

All things considered, the 555 chip is an extremely widely integrateable chip that serves its purpose reliably and efficiently. The results from all 4 of my experiments were nearly exactly as expected, with a certain degree of error that was also expected and the source of which has been identified. It is fair to say that the experiment is well designed too, but could be improved by providing a streamlined datasheet. Unless reading and interpreting datasheets is an intended purpose of this task, there is a large sum of information that is cumbersome to navigate and unuseful during these experiments which leads to quite a lot of confusion and wasted time. Overall though, this project clearly outlines the function of a 555 timer, and exposes the wide array of potential uses for it, while also demanding the engagement of the creative side of the brain to invent a new circuit, which is what engineering is all about.

The goals for this experiment were met, the methods weren’t straightforward (but they arent meant to be), but they were extremely insightful , and the objectives were all met to a high degree of accuracy.

6. References

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